Establishing a Trustworthy Accessibility Baseline: A WCAG-Driven Checklist for Auditable Practice in Digital Projects

Abstract

In the digital space, it is important to ensure that every person, with or without disabilities, can access and engage with online content. Yet organizations that build digital solutions do not always turn this commitment into a practice they can check. This practice-based study, developed in a Brazilian research and development institute, asks how an organization handles accessibility guidelines in practice, how this compares with the market, and what it means for trust between digital products and their users. It began by mapping practice through a self-reported survey of 42 projects, 41 never audited. Because maturity was low, Level A of the Web Content Accessibility Guidelines (WCAG) was proposed as an auditable baseline, and a checklist was built for designers, developers, and quality assurance (QA) teams. Its UI/UX subset, applied to a pilot project through an AI-assisted evaluation still under human review, made basic failures visible. The findings, which market surveys echo, suggest that what is missing is less accessibility work itself than a way to verify it. Read through trust theory, this matters because an interface that fails basic criteria signals that some people who could use it were not considered. We argue that WCAG is not merely a technical checklist but a design act of trust-building, with ethical and social weight, when it turns commitment into everyday, verifiable practice. The study offers a way of reading inclusive design through a trust lens, and a first step, for UX and development teams, from commitment toward auditable practice.

Keywords: Digital Accessibility; WCAG; Trust; Practice-Based Research; Auditable Practice.

1. Introduction

Digital accessibility has become part of everyday product development. This growing recognition, and the practical question of where to begin structuring it, gave rise to this study. The formal research question follows from there: how have organizations, and the agents involved in building digital products, engaged with accessibility guidelines in practice, and what does the presence or absence of that engagement mean for the trust relationship between these products and the people who use them?

This concern starts from a simple fact: digital services have become central to how life is organized. Internet access, for example, enables the use of a wide range of public and private

services. In the European Union, for instance, Kużelewska et al. (2025) report that in 2024, 70.01% of citizens used e-government services and 39.78% used e-health services, besides common uses for internet-based communication, online news access, and entertainment sites.

Yet this expansion of our digital life does not reach everyone on equal terms. According to the World Health Organization (2023), approximately 1.3 billion people, around 16% of the global population, experience significant disability. Disability is part of human experience and can affect any person at any stage of life, temporarily or permanently. Given this reality, it is necessary to ensure that online services offer accessible interfaces.

Our research question dialogues with the track Design as Social Production: Trust, Distrust, and the Legitimation of Social Realities. When an interface fails to meet baseline accessibility requirements, it signals users with disabilities and other excluded groups that they were not considered in the design process, undermining the trust between institutions and the people they serve.

In this study, we investigated a Brazilian research and development institute that builds digital solutions for worldwide clients. In practice, it operates much like a software house, serving clients across sectors such as agribusiness, healthcare, logistics, and others, navigating diverse contexts and regulatory requirements.

Beyond this diversity of contexts, the organization is also shaped by broader global themes, among them ESG (Environmental, Social and Governance), institutionalized since 2022. Across its annual ESG reports through 2025, “accessibility” has found its place and remained in these documents within the Diversity, Equity and Inclusion guidelines, through social actions such as sign language interpreters and video captions. As the organization's ESG governance matured, with reporting aligned with international frameworks, the theme of accessibility remained socially framed but technically undefined, without dedicated goals, audit indicators, or references to technical standards for digital solutions. Efforts, however, have begun to shift with this research. Accessibility starts to move from social action toward a technical position, through practices that can be verified in the development of digital solutions. Reading that movement requires looking at how commitment, individual knowledge, and trust relate to verifiable practice.

2. Literature Review / Background

Accessibility can arise from a social commitment and from the initiative of individual people, before it becomes something an organization can examine. Moving from one to the other is a

design problem in which stakeholders hold different goals. This study approaches it through practice-based research, in line with research through design (Zimmerman et al., 2007). In this approach, designers work from within the situation and produce artifacts to generate knowledge, not only to apply it. These artifacts move the situation from its current state toward a preferred one, and they become exemplars through which findings reach research and practice communities. Researchers must also state what the preferred state is. Here, it is accessibility that an organization can examine and verify.

One line of prior work points to individual developers. In Brazil, Silva and Rodrigues (2018) ran an exploratory survey with 25 programmers and tested time, budget, interest, and knowledge as possible causes of inaccessible systems. They found that low knowledge of accessible programming techniques was the main one, and they suggested training as a response. The finding concerns developers' technical knowledge, not design knowledge in general. Individual knowledge matters, but it does not by itself make accessibility expected, measurable, and verifiable across teams. Where it is not verifiable, the consequences fall on the people who use the products.

Trust offers a way to read what is at stake for them. Mayer, Davis, and Schoorman (1995) define trust as the willingness to be vulnerable to another party, based on the expectation that this party will act in ways that matter to the trustor, regardless of the ability to monitor it. That expectation rests on the perceived ability, benevolence, and integrity of the trustee. Lee and See (2004) bring the idea to technology and argue that reliance is appropriate when trust is calibrated to what a system actually does. Calibration needs evidence of what a system does, and verifiable practice is a source of that evidence. Neither work addresses accessibility. Still, users rarely see how a product was built, so they read the interface itself. An interface that fails basic criteria signals who was considered, and whether the institution behind it can be relied upon.

Verifiable practice, in turn, requires a shared reference, and the Web Content Accessibility Guidelines (WCAG), maintained by the W3C, provide one. They organize accessibility into four principles, known as POUR. Content must be Perceivable, Operable, Understandable, and Robust (W3C, 2023). Conformance has three levels, A, AA, and AAA. Level A holds the most fundamental requirements, and each level builds on the one before it. WCAG is supported by documented techniques for meeting its criteria, which speaks to the knowledge gap reported above. It is a technical standard, not a full definition of accessibility, which is a broader and ongoing commitment.

3. Methodology / Approach

As a first step, this study mapped the current state of accessibility practice across projects using an 11-item questionnaire, detailed in Appendix A, covering the technical stack, design system, and accessibility tools adopted. The questionnaire was sent to 58 active projects and answered by 42 (72%) between 18 and 25 June 2026, and responses were self-reported. The full study is composed of four steps, illustrated in Figure 1.

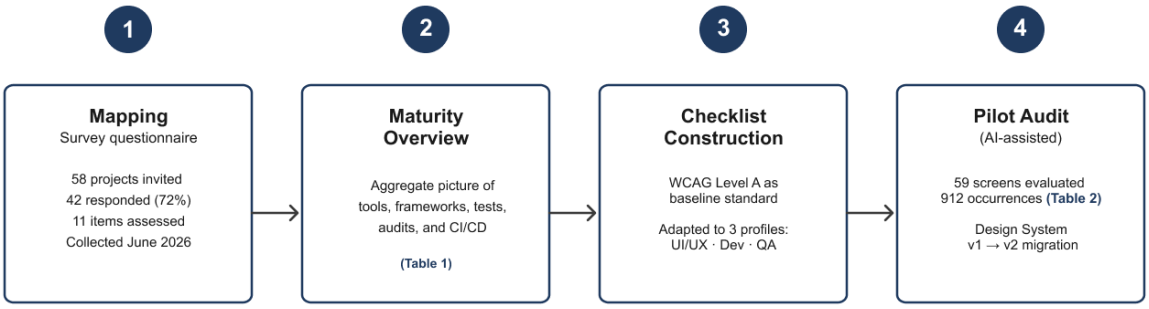


Figure 1. Four-step methodological flow: mapping, maturity overview, checklist construction, and pilot audit.

The resulting maturity overview (see Table 1 below) revealed that maturity was still insufficient to support more advanced criteria, which led to proposing WCAG 2.2 Level A as the baseline for all projects in the next step. Building on that baseline, a checklist was developed and tailored to the institute's product development pipeline for the main development agents (UI/UX designers, developers, and QA analysts), a precedent already seen in Brazil's e-MAG checklist (Federal Government, 2010). This way, each area has its own criteria for any new feature, from design through test validation.

The pilot project selected for the checklist application (step 4, Figure 1) was undergoing migration from version 1 to version 2 of its Design System at the time of data collection, still coexisting with legacy components that had not been updated. Data collection was carried out through an AI-assisted evaluation of the project's design screens through the checklist, producing the WCAG 2.2 Level A audit records that serve as this stage's primary data source. The AI-assisted results are treated as provisional and are still being cross-checked through human review.

The research design follows the logic of practice-based research, discussed in the Literature Review. The mapping survey, the checklist, and the pilot application were part of an iterative process, in which the knowledge jointly built by the researchers and collaborators shaped both

the research design and the checklist criteria. In this sense, the checklist and the maturity survey themselves function as artifacts of knowledge produced throughout the investigation, not as neutral instruments of data collection.

4. Findings / Outcomes

The accessibility maturity survey evaluated 42 active projects across four dimensions: tools, framework and WCAG compliance, automated testing, and CI/CD pipeline integration (Table 1). The same pattern recurs across all four dimensions: formal practice is almost entirely absent, even where the supporting infrastructure already exists. Only 4 of the 42 projects report applying WCAG Level AA. The clearest example of this gap is CI/CD. 81% of projects already have this pipeline in place, but none of them include automated accessibility verification.

Table 1. Accessibility maturity overview across the projects investigated (N = 42)

Dimension	**Finding**	**n**	**%**
Accessibility tools	No tool, or status unknown	33	79%
	Uses some tool	9	21%
Accessibility framework adopted	None, or unknown	28	67%
	Formal reference to WCAG	11	26%
	Best practices, ARIA, or other	3	7%
WCAG compliance level applied	None, or unknown	31	74%
	Level A	7	17%
	Level AA	4	10%
Automated testing and auditing	No automated testing	42	100%
	Never audited	41	98%
	Recent audit reported	1	2%
CI/CD pipeline	Pipeline implemented	34	81%
	Pipeline includes accessibility checks	0	0%

As the study's second artifact, the design checklist audit, applied to the pilot project, deepened these findings beyond the maturity survey's aggregate indicators. The full checklist covers designers, developers, and QAs, but only the 27-item subset built for UI/UX (Appendix B) was applied at this stage, evaluated across 59 prototype screens of the pilot project. Each occurrence audit corresponds to one checklist item applied to one specific screen. Items that do not apply to a screen were grouped into a single N/A record for that screen. The full breakdown of the 912 occurrences recorded appears in Table 2.

Table 2. Distribution of occurrences in the checklist audit (design pilot, 59 screens, N = 912)

Occurrence status	**Definition**	**n**	**%**
Failed	Does not meet the accessibility criteria evaluated. Of these 339 failures, 148 (43.7%) originated in the Design System itself and 26 (7.7%) from incorrect use of DS components, together 174 (51.3%).	339	37.2%
Attention needed	Partial compliance, or a design decision still pending	289	31.7%
Compliant	Meets the accessibility criterion evaluated	90	9.9%
Flagged as dev requirement	Outside design scope, depends on technical implementation	134	14.7%
Not applicable	Items that do not apply to the screen, grouped per screen with justification	60	6.6%

As Table 2 shows, more than half of all failures (51.3%) originate in the Design System itself or from the incorrect use of its components. Failures originating in the Design System are fixable once in the shared library, rather than screen by screen. As described in the Methodology, many screens also use components built locally instead of library components, which adds inconsistencies screen by screen. In this pilot, the pattern suggests that most accessibility problems do not stem from isolated, screen-level decisions, but from gaps in the component library and in how it is used.

Taken together, both artifacts point in the same direction. Some projects report applying WCAG, but almost none can verify it, since 41 of the 42 were never audited, and none ran

automated checks. These findings suggest that the issue is less the absence of practice than the absence of verification. This is where the question of trust begins. Without a verifiable signal that basic criteria were met, users have little way of knowing whether they were considered.

5. Discussion

5.1 Interpreting the Findings

The data gathered indicate that most projects have no verifiable record of their own accessibility state. In this scenario, failures may surface only when they reach the people who use the product. This scenario dialogues with broader digital accessibility panoramas, the Brazil-Portugal Panorama (Sales & Ferreira Duarte, 2026) and the HandTalk/MWPT (2023) survey, summarized in Table 3, which trace a movement from recognition to practice. Recognition is broad, as 51% of HandTalk/MWPT (2023) respondents name the inclusion of people with disabilities as the main focus of their projects for the year. Practice is thinner. In the Brazil-Portugal Panorama (Sales & Ferreira Duarte, 2026), legal requirement is the most cited investment motivator, the most common perception is that enforcement occurs only in specific cases or relevant complaints, and about a quarter of respondents report regular evaluations by specialists. Compliance checking thus tends to be reactive in the market, and the absence of audits and automated tests in the surveyed projects points in the same direction. Although these surveys are not representative samples, they suggest that the absence of systematic measurement is not exclusive to the organization studied.

Table 3. Convergence across studies: from recognition to practice in digital accessibility

Source	**Indicator**
Sales and Ferreira Duarte (2026), 111 respondents (Brazil and Portugal).	79.3% cite legal requirement as a motivator for investing in accessibility.
	45% believe enforcement occurs only in specific cases or relevant complaints
	28.8% believe legislation is not enforced in their country
	26.4% report regular evaluations by accessibility specialists (at least monthly)
HandTalk / MWPT (2023), 181 respondents	51% rank inclusion of people with disabilities as the social cause their company's projects focus on this year

from companies with digital accessibility initiatives.	79% rank the Social pillar of ESG as a top priority
	Only 33% report hiring digital accessibility specialists

5.2 Accessibility as a Design Act of Trust

Blanck (2014) argues that when industry self-regulation and market incentives fail to protect people with disabilities, prescriptive standards become necessary to prevent paternalistic practices, unfair treatment, non-consensual data use, and exclusion, particularly for those with the least power to demand equal digital access (p. 25). Adopting good accessibility practices, especially through a compliance standard such as WCAG, turns a technical specification into an instrument of social and ethical accountability in the digital space.

Trust with communities of users with disabilities and other excluded groups is not established when those users are absent from the design, development, and delivery of digital products. Inclusive products require awareness of diverse human needs and an active organizational commitment to embed accessibility in every stage of the product lifecycle. Audits, tools, and automated testing make that commitment measurable and verifiable, even if adoption alone does not guarantee its full realization. Read through Mayer, Davis, and Schoorman (1995), verifiable practice offers evidence of integrity. Read through Lee and See (2004), it allows trust to be calibrated to what the product does.

5.3 Implications for the Organizational Process

The checklist represents a first institutional approach to integrating accessibility into the existing product/project lifecycle. The pilot's AI-assisted evaluation suggests that a first-pass audit is feasible, although its results remain provisional and depend on human review. The CI/CD finding points the same way. The barrier appears to be organizational rather than technical, since the infrastructure to close that gap already exists. Adopting the checklist as an institutional instrument, rather than a one-off application, speaks to the distance between the accessibility actions already present in the organization's ESG agenda (internal ESG reports since 2022) and technical accessibility practice, as discussed in the Introduction. Table 3 suggests that this gap is not exclusive to the organization studied. HandTalk/MWPT (2023) pairs high priority for the ESG Social pillar with limited specialist hiring, and Sales & Ferreira Duarte (2026) point to reactive enforcement and limited specialist review.

The regulatory landscape reinforces this urgency. In the European Union, the European Accessibility Act (Directive (EU) 2019/882) has required many digital products and services,

such as e-commerce and banking, to be accessible since June 2025 (European Parliament & Council of the European Union, 2019). In Brazil, ABNT NBR 17225:2025 (ABNT, 2025) sets technical requirements for web accessibility, and Bill No. 2251/2025 (Brazil, Federal Senate, 2025), still in progress, would institute the National Digital Accessibility Day. Its justification cites a survey by BigDataCorp and Movimento Web para Todos in which only 2.9% of Brazilian websites passed every accessibility test in 2024. In other words, even where standards exist, practical adoption remains low. This is the paradox at the core of this study, in which norms exist while practice remains unevenly institutionalized. Level A is a starting baseline. It sits below the level that the standard treats as regular conformity, which covers Levels A and AA of WCAG 2.2.

In this context, turning the checklist into an ongoing institutional process moves the organization closer both to emerging regulatory compliance and to a more substantive reading of its own ESG commitment. The organization already acts on Diversity, Equity, and Inclusion, but does not yet have auditable technical accessibility practice in its development pipeline. The checklist is the first proposal to narrow this distance. It expands accessibility beyond D&I actions toward an accessibility policy, which Kalbag (2017) describes as a set of guidelines that are clear, hierarchical, and testable.

5.4 Limitations

The study is specific to a single organization, and the survey covers only the 42 of 58 invited projects that responded. The indicators capture only formal, measurable practices, such as tool adoption, WCAG reference, and CI/CD integration, so informal accessibility work in some teams was not visible. The survey responses are also self-reported, and the study includes no data from users, so its claims about trust are interpretive. In addition, the checklist was applied to a single pilot project, and only to its design (UI/UX) item subset. The audited occurrences are still under human review, and corrections have not yet reached the production version.

A conceptual limitation also deserves acknowledgment. Formalizing into a checklist an accessibility knowledge that was previously implicit and instinctive risks reducing the contextual sensitivity it sought to capture. A compliant item shows that a criterion was checked, not that a genuinely more accessible experience was considered.

Even so, this study contributes directly to the conference theme on trust and distrust in design. It shows that accessibility standards can exist without the practice being institutionalized in an

organization. It therefore argues that trust between organizations and their users, including people with disabilities and other excluded groups, does not come from the existence of standards or policies. It comes from the everyday, verifiable practice those standards produce. In this sense, the Design System examined in the pilot functions as a concrete case of how practices, beliefs, and power relations distributed across multiple agents (designers, developers, and component-library maintainers) sustain or contest the legitimacy of an inclusive digital experience.

6. Conclusion

This study sets out to understand how organizations and the agents who build digital products engage with accessibility guidelines in practice. Across the 42 projects, formal engagement with these guidelines is largely absent. Accessibility is not accounted for, measured, or tested in a verifiable way throughout the product lifecycle, and in the pilot audit, more than half of the failures are tied to the Design System and its use. This has implications for the trust relationship between these products and the people who use them.

Proposing WCAG Level A as the organizational baseline is a first step toward turning a commitment to the people who use these products into practice that can be verified, and toward extending the accessibility actions already present in the organization's ESG agenda to its digital products.

Committing to measure, test, and integrate accessibility into the workflows, as an ongoing commitment rather than a project that gets finished (Byrne-Haber, 2021, p. 32), is to include everyone, and inclusion is a matter of trust. This work understands that accessibility standards carry ethical weight (Blanck, 2014), and that building accessible products matters for building trustworthy products.

Future work includes formalizing the checklist based on WCAG with Level A as an initial internal standard, conducting accessibility audits for projects that have never had one, and integrating automated accessibility checks into the CI/CD pipelines that are already in place. This turns existing infrastructure into a foundation for more inclusive digital products.

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Appendix

A. Accessibility Maturity Survey Instrument

#	Type	Field	Response options
1	Identification	Project type	Software / Website / Internal product / Mobile app / Other
2		Current status	In development / In maintenance / Completed
3		Point of contact for accessibility matters	Name and role / N/A
4	Technical context	Technical stack/framework	Free text (e.g., React, Angular, Vue)
5		Design system	Free text (e.g., Ant Design, Material Design)
6		Accessibility tool	Lighthouse / Axe / Other / None / Unknown
7		Accessibility framework	WCAG / Other / None / Unknown
8		WCAG conformance level applied	A / AA / AAA / None / Unknown
9		Has CI/CD pipeline	Yes / No

10		Automated accessibility testing	Yes / No
11	Process	Last accessibility audit	Never / More than a year ago / Recently

Note. The "Point of contact" field is administrative, used for the research team's internal follow-up, and does not produce data analyzed in this article.

B. WCAG 2.2 Level A Accessibility Checklist (27 items, UI/UX)

27-item subset applied in the pilot, derived from the full accessibility checklist (which also covers developers and QA).

#	**Principle**	**WCAG Criterion**	**Item verified**
1	Perceivable	1.1.1	Every image, icon, or captcha has defined alternative text
2	Perceivable	1.2.1/1.2.2/ 1.2.3	Pre-recorded video or audio has captions, transcript, or audio description
3	Perceivable	1.3.1/1.3.2	The visual reading order also makes sense to screen reader users
4	Perceivable	1.3.3	No instruction relies solely on color, position, or sound (e.g., "click the green button")
5	Perceivable	1.4.1	Color is not the only indicator of error, status, or action (always paired with icon/text)
6	Perceivable	1.4.2	Auto-playing media has controls to pause, mute, or adjust volume
7	Operable	2.1.1/2.1.2	Interactive elements are keyboard-accessible, with visible focus
8	Operable	2.4.3	Tab order is defined and logical
9	Operable	2.1.4	Keyboard shortcuts do not rely only on unhandled single letters/numbers
10	Operable	2.2.1/2.2.2	Time-limited actions can be extended, paused, or disabled

11	Operable	2.3.1	No element flashes more than 3 times per second
12	Operable	2.4.1	A way to skip repetitive blocks exists (skip link)
13	Operable	2.4.2	Every screen has a clear, descriptive title
14	Operable	2.4.4	Links and buttons make sense out of context (avoid "click here")
15	Operable	2.5.1	Complex gestures (drag, pinch) have a simple alternative
16	Operable	2.5.2	An accidental click or tap can be cancelled before confirmation
17	Operable	2.5.3	The visible label matches, or is compatible with, the component's accessible name
18	Operable	2.5.4	Motion-activated functions have a touch-based alternative
19	Understandable	3.1.1	The screen/content language is defined and documented
20	Understandable	3.2.1/3.2.2	No context change happens automatically without user action
21	Understandable	3.2.6	Help patterns keep the same format across all screens
22	Understandable	3.3.1	Errors are identified with color, icon, and a text message
23	Understandable	3.3.2	All form fields have clear labels and instructions
24	Understandable	3.3.7	Multi-step forms do not repeat the same request for information
25	Robust	4.1.2	Custom components have name, role, and value specified for ARIA
26	Process	N/A	Accessibility annotations are included in the design file

27	Process	N/A	Non-applicable items are marked N/A with justification